AIRBUS

ABS 5455

Issue 3
Page 1 of 7
January 2006

Aerospace series
Round bars
Precipitation Hardening Stainless Steel (X5CrNiCu15-5(15-5PH))
Normal and special diameter tolerances (codes N and S)
Diameter 6,0 mm < D ≤ 250,0 mm

Published and distributed by :
AIRBUS S.A.S.
ENGINEERING DIRECTORATE
31707 BLAGNAC Cedex
FRANCE

Ref. no. ABS5455

1 Scope

This standard specifies the dimensions and tolerances of round bars in precipitation hardened stainless steel (X5CrNiCu15-5), diameter 6 mm < D ≤ 250 mm for aerospace applications.

2 Normative references

This Airbus Standard incorporates by dated or undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus Standard only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies (including amendments).

EN 3848	Aerospace series; semi finished metallic products; methods of measuring form deviations. [1]
AIMS 01-04-000	Technical specification - Steel wrought products - Plate, sheet, strip, bar and section.
AIMS 01-04-003	Material specification – Precipitation hardened stainless steel (X5CrNiCu15-5(15-5PH)) – Round bars – Rm ≥ 1070 MPa – Diameter 6,0 mm < D ≤ 250,0 mm

3 Form

See Figure 1.

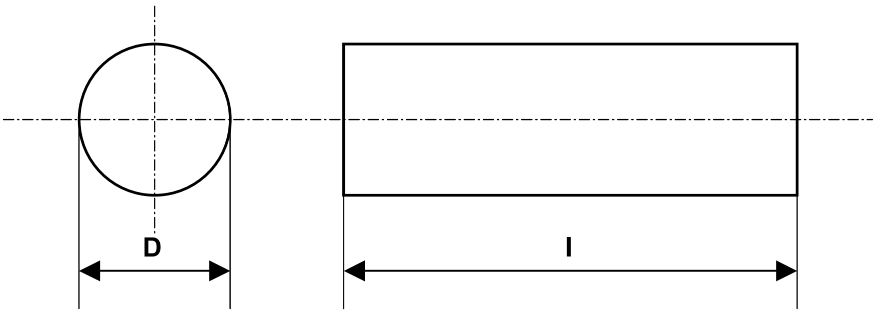

Figure 1

[1] Published as AECMA Prestandard at the date of publication of this standard

4 Recommended dimensions

See Table 1.

Table 1

Dimensions in mm

Size code	D[1)] Nominal Diameter	l length
0070	7	The length shall be specified in the order.
0080	8	
0090	9	
0100	10	
0200	20	
0300	30	
0400	40	
0500	50	
0600	60	
0700	70	
0800	80	
0900	90	
1000	100	
1200	120	
1500	150	
1700	170	
2000	200	
2300	230	
2500	250	

1) Other diameters on request
2) for information, density is 7,85 kg/dm^3

5 Tolerances

5.1 Dimensional tolerances

5.1.1 Diameter

5.1.1.1 Normal diameter code N

See Table 2.

Table 2

Dimensions in mm

Diameter	Normal tolerances
6 < D ≤ 15	± 0,3
15 < D ≤ 25	± 0,4
25 < D ≤ 35	± 0,5
35 < D ≤ 50	± 0,6
50 < D ≤ 80	± 0,8
80 < D ≤ 100	± 1,0
100 < D ≤ 120	± 1,3
120 < D ≤ 160	± 1,6
160 < D ≤ 200	± 2,0
200 < D ≤ 250	± 1,2% of diameter

5.1.1.2 Special diameter code S

See Table 3.

Table 3

Dimensions in mm

Diameter	Special tolerances
6 < D ≤ 12	± 0,15
12 < D ≤ 18	± 0,18
18 < D ≤ 30	± 0,21
30 < D ≤ 50	± 0,25
50 < D ≤ 80	± 0,30
80 < D ≤ 120	± 0,50
120 < D ≤ 250	± 1,00

5.1.2 Length

See Table 4

Table 4

Dimensions in mm

Length	Tolerances
All	+ 6 0

5.2 Geometric tolerances

5.2.1 Straightness

5.2.1.1 Method of measurement and symbols

See EN 3848.

5.2.1.2 Tolerances

See Table 5

Table 5

Dimensions in mm

Diameter	Maximum Straightness Deviation	
	Y_1 per meter	Y_2 on any length X_2 = 400 mm
6 < D ≤ 250	1	0,5

5.2.2 Roundness

5.2.2.1 Method of measurement and symbols

See EN 3848.

5.2.2.2 Tolerances

See Table 6
The lateral curvature may be concave or convex.

Table 6

Dimensions in mm

Diameter	Maximum Roundness Deviation
6 < D ≤ 12	0,15
12 < D ≤ 18	0,18
18 < D ≤ 30	0,21
30 < D ≤ 50	0,25
50 < D ≤ 80	0,30
80 < D ≤ 120	0,50
120 < D ≤ 250	1,00

6 Material

See Table 7

Table 7

<table>
<tr><th colspan="8">Material</th></tr>
<tr><th colspan="4">Delivered</th><th colspan="3">Use</th></tr>
<tr><th>AIMS spec. No.</th><th>Condition of treatment</th><th>Code</th><th>Diameter tolerance codes</th><th>AIMS spec. No.</th><th>Condition of treatment</th><th>Code</th></tr>
<tr><td rowspan="2">01-04-003</td><td>Solution treated + precipitation hardened*</td><td>A</td><td rowspan="2">N or S</td><td rowspan="2">01-04-003</td><td>Solution treated + precipitation hardened</td><td>A</td></tr>
<tr><td>Solution treated**</td><td>B</td><td>Solution treated + precipitation hardened by the end user *</td><td>C</td></tr>
</table>

* solution treated + precipitation hardened = .4 according to DIN designation
** solution treated = .9 according to DIN designation

7 Designation

7.1 Standard designation

EXAMPLE:

	Description block	Identity block
	Round Bar	ABS5455 A 007 (N)
Number of this standard		ABS5455
Material code in acc. with Table 6		A
Size code in acc. with Table 1		007
Diameter tolerance codes in acc. with paragraph 5.1.1, for procurement only		(N)

8 Technical specification

AIMS 01-04-003.

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
Issue 1 03/04		New Standard.
Issue 2 02/05		Modification of maximum diameter
Issue 3 01/06		Modification of diameter tolerances